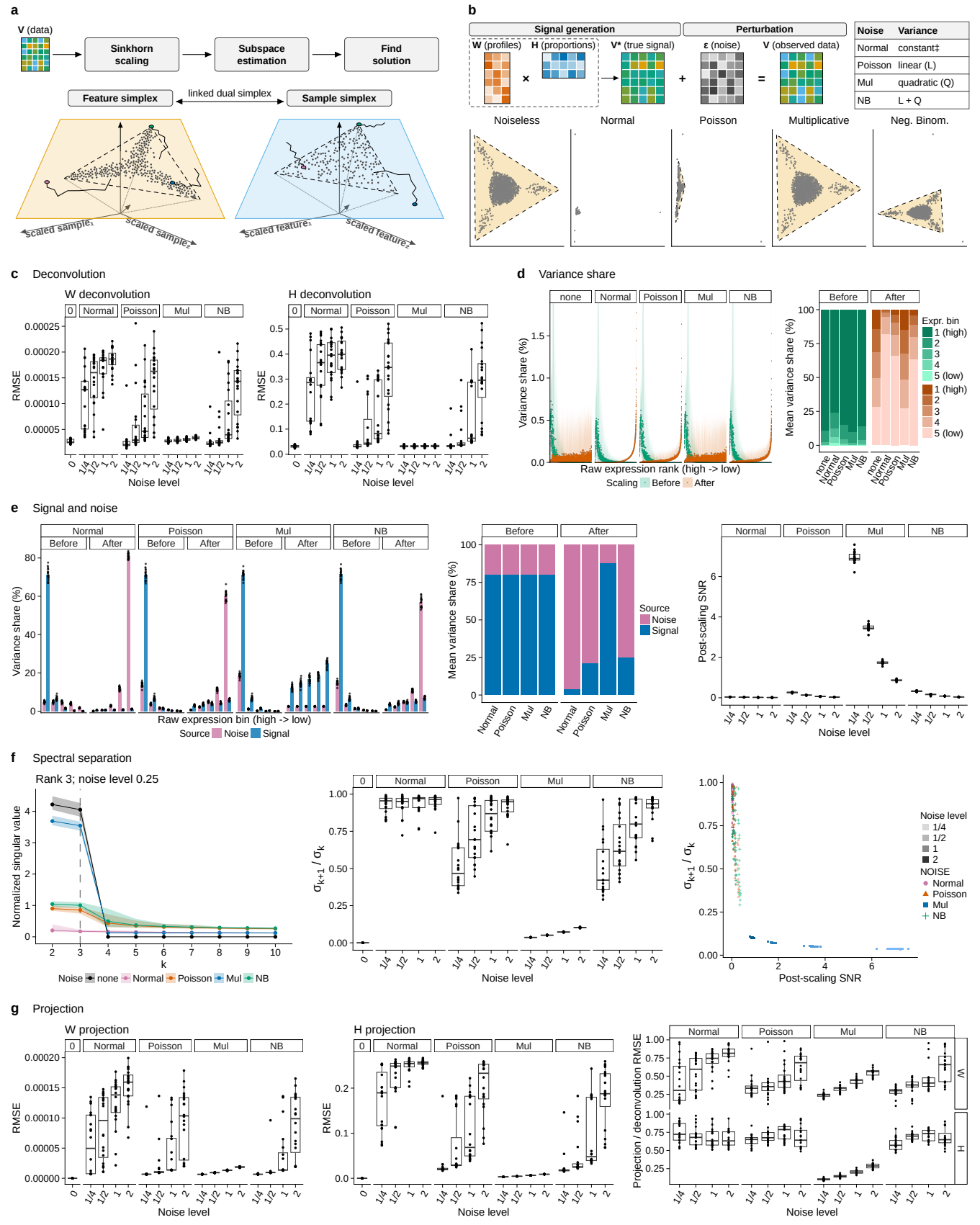

a
V (data)
Sinkhorn scaling
Subspace estimation
Find solution
Feature simplex
linked dual simplex
Sample simplex
Scaled sample₁
Scaled sample₂
Scaled feature₁
Scaled feature₂
b
Signal generation
Perturbation
W (profiles)
H (proportions)
V* (true signal)
ε (noise)
V (observed data)
Noise | Variance
Normal | constant‡
Poisson | linear (L)
Mul | quadratic (Q)
NB | L + Q
Noiseless
Normal
Poisson
Multiplicative
Neg. Binom.
c
Deconvolution
W deconvolution
H deconvolution
RMSE
Noise level
d
Variance share
Variance share (%)
Raw expression rank (high -> low)
Scaling Before After
Mean variance share (%)
Expr. bin
1 (high)
2
3
4
5 (low)
e
Signal and noise
Variance share (%)
Raw expression bin (high -> low)
Source Noise Signal
Mean variance share (%)
Post-scaling SNR
Noise level
f
Spectral separation
Rank 3; noise level 0.25
Normalized singular value
k
Noise none Normal Poisson Mul NB
σ_{k+1} / σ_k
Post-scaling SNR
Noise level
NOISE
Normal
Poisson
Mul
NB
g
Projection
W projection
H projection
RMSE
Noise level
Projection / deconvolution RMSE
W
H